



MINHAJ ISLAMIC MODEL SCHOOL HAZRO
FIRST TERM EXAMINATION (2026)

Class: 10th Subject: Physics Total Marks: 50

Roll No: _____ Student Name: _____ Time Allowed: 2 Hours

SECTION A: OBJECTIVE TYPE (10 Marks)

Q1. Choose the correct option for each question:

(i) When the temperature of a rod of copper is increased, its length:

[A] remains the same [B] increases [C] decreases [D] becomes double

(ii) What does a liquid turn into once it is evaporated?

[A] Gas [B] Water vapours [C] Air [D] Both (a) and (b)

(iii) High-temperature superconductors typically involve which type of material?

[A] Pure metals [B] Organic compounds [C] Ceramics [D] Liquid crystals

(iv) Which method of heat transfer requires direct contact between particles?

[A] Convection [B] Radiation [C] Conduction [D] Evaporation

(v) The natural phenomenon caused by convection currents is:

[A] earthquakes [B] land and sea breezes [C] magnetic fields [D] solar eclipses

(vi) How does heat transfer occur through radiation?

[A] Through direct contact between objects [B] Through the movement of heated fluid [C] Through electromagnetic waves without needing a medium [D] none

(vii) The Sun's heat reaches the Earth:

[A] through conduction between air particles [B] through convection in space [C] through radiation via electromagnetic waves [D] through reflection by the moon

(viii) Which type of surface is the best absorber of radiation?

[A] smooth and shiny surfaces [B] dark and rough surfaces [C] transparent and thin surfaces [D] polished and white surfaces

(ix) Why does a thermos flask have a vacuum between its double walls?

[A] To allow conduction and convection [B] To stop heat transfer by conduction and convection [C] To increase heat absorption by radiation [D] To keep the outer casing hot

(x) Which factor increases the rate of evaporation?

[A] Decreasing temperature [B] Increasing surface area of the liquid [C] Increasing humidity [D] Decreasing wind speed

SECTION B: SHORT ANSWER QUESTIONS (10 × 3 = 30 Marks)

Q2. What factors influence the thermal expansion of solids?

Q3. What is latent heat of fusion?

Q4. How does evaporation contribute to cooling processes in everyday life?

Q5. What is latent heat of vaporization, and how is it calculated?

Q6. What are the three primary types of heat transfer?

Q7. Why does heat travel faster in metals compared to non-metals?

Q8. What natural phenomenon causes land and sea breezes?

Q9. What is radiation, and how does it transfer heat?

Q10. Why does a dull black surface emit and absorb more radiation than a shiny silver surface?

Q11. Give the applications of Heat transfer

SECTION C: DESCRIPTIVE / LONG QUESTIONS ($2 \times 5 = 10$ Marks)

Attempt any TWO (2) questions. Each question carries 5 marks.

Q12. Compare the absorption and emission of heat by different surfaces. How does the colour and texture of a surface affect its ability to absorb and emit heat?

Q13. What is Evaporation? Give factor affecting the rate of Evaporation.